Data Cleaning Report

Dataset Name: Depaul_dataset (International Student Applications Dataset)

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Introduction

The Data cleaning process was carried out to prepare the International Student Applications Dataset for analysis, to ensure data accuracy, completeness and consistency. The dataset contains 7544 rows including header, and 80 columns.

Overview of the Data Quality Issues

Issue Type	Description	Columns Affected	Impact
Empty Columns	Some columns had no values in it	'Major', 'Degree_Type', 'Counsler', 'Citizenship', 'Status', 'Major_1st_Choice', 'Street_3', 'Date_of_Birth', 'Phone_Number_2', 'Most_Recent_Released_Decision', 'RIT_Email_Created', 'Housing_Contract', 'Is_Global_Grad', 'Is_Admitted', 'SEVIS_ID', 'Official_University_email_address', 'Application_Agency_Code', 'Outstanding_Checklist_Items', 'Prior_I-20_Outreach_Detail', 'Prior_Non_I-20_Outreach_Detail', 'Most_Recent_Contact', 'Most_Recent_User_and_Date', 'Recent_GT_Form_Initiative', 'Caller_Name_2', 'Date_of_Contact_2', 'Outcome_2', 'Final_Result', 'Escalation_Required', 'SLU_Start_Comment', 'SLU_Start_City', 'I_901_Status', 'Category', 'Intake-2', 'Status-2', 'City_and_Branch', 'Header', 'Template', 'Region-2'	No meaningful impact on analysis.
Missing Values	Some columns had less or more	'Given_Name', 'College', 'Country', 'Intake', 'College_1st_Choice', 'Phone_Number', 'Street_1', 'Street_2', 'City', 'Region', 'Postal',	Reduced Completeness

	than 80% missing data	'Email_ID', 'Comments', 'Admit_Date', 'Reference_ID-2', 'Recieved_At-2', 'University-2', 'Date_of_Contact', 'Caller_Name', 'Outcome_1', 'Remark', 'Remark_2', 'Campaign_Id', 'Slate_Form_Filled', 'Created_At-2', 'Modified_At-2', 'ID', 'Name', 'Attempts', 'University-3', 'Degree_Type-2', 'Type', 'Start_Date', 'Created_At-3', 'Modified_At-3'	
Duplicates	A few duplicate records found	Reference_ID	Risk of bias
Data Type Errors	Date stored as text	Recieved_At-2, Admit_Date, Date_of_contact	Wrong data type
Outlier	No outlier found	Nil	May distort analysis

Data Cleaning Steps and Methods

Cleaning Steps	Description	Python method/code used	Before	After
1.Handling Empty Columns	Dropped columns with no values(empty columns) in it.	df=df.dropna(axis=1, how='all')	80 columns	42 columns

2. Handling Missing Values	Dropped 2 columns with >90% missing data (Remark, Remark_2); filled numerical columns with mean and others with mode.	[for col in df.select_dtypes(include='object'): df[col].fillna(df[col].mode()[0], inplace=True)] [for col in df.select_dtypes(include='datetime'): df[col].fillna(df[col].mode()[0], inplace=True)] [df.fillna(df.mean(numeric_only=True), inplace=True)]	42 columns	15 columns
3. Removing Duplicates	Removed all duplicate rows	df.drop_duplicates('Reference_ID',inplace=True)	7543 records 2543 duplicates	5000 records 0 duplicate
4. Fixing Data Types	Converted text dates to datetime format	df['Admit_Date']= pd.to_datetime(df['Admit_Date']) df['Recieved_At-2']= pd.to_datetime(df['Recieved_At-2']) df['Date_of_Contact']= pd.to_datetime(df['Date_of_Contact'])	Object64	Datetime64

Post-Cleaning Summary

Metric	Before Cleaning	After Cleaning
Rows	7543	5000
Columns	80	40
Missing Values	81% columns	0% columns
Duplicates	2543	0
Invalid Data Types	3	0

After cleaning, the dataset now contains 5000 Valid records with no missing values, consistent data types, and no duplicates. It is ready for analysis and visualization.

Insights from the Cleaning Process

1. Columns with >90% missing values were removed to maintain data integrity.
2. Data type issues were resolved; all dates are now in datetime format.

Recommendations

Issues	Recommendation
Missing values	Improve data collection and consistencies.
Duplicates	Use Reference ID as unique constraint in future data
Data type errors	Validate formats at data entry stage
Outliers	Add range checks during data input

Visualization

Before cleaning and After cleaning

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5]: <Axes: ylabel='Reference_ID'>
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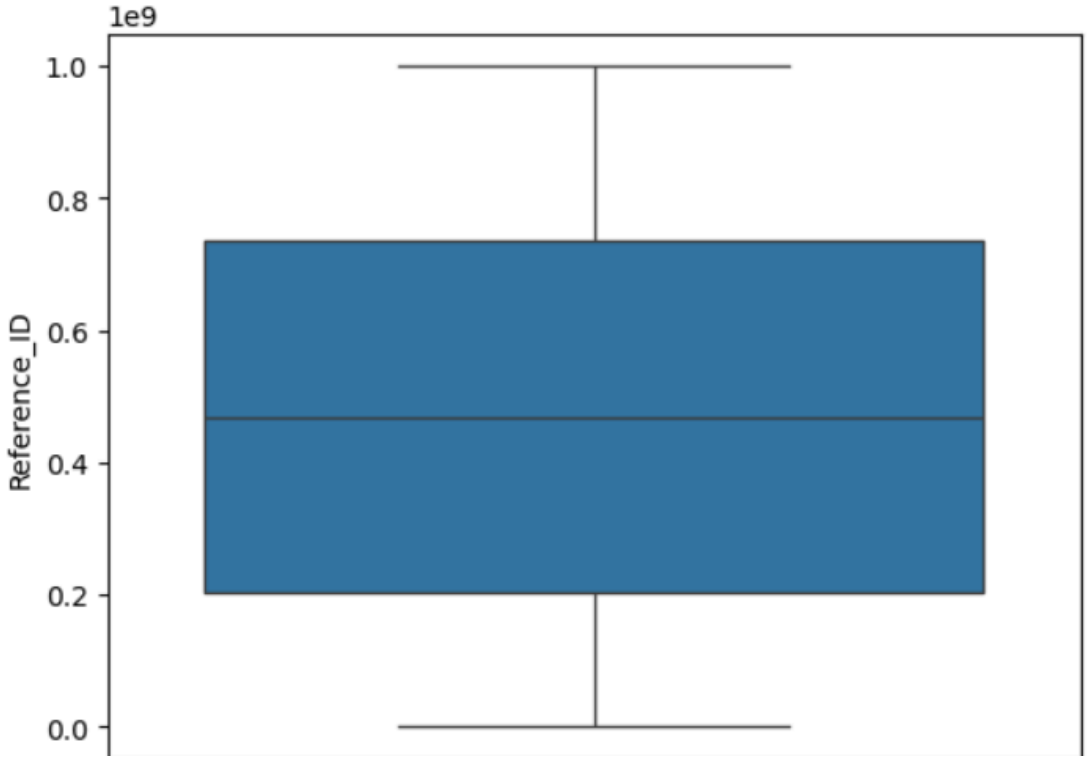
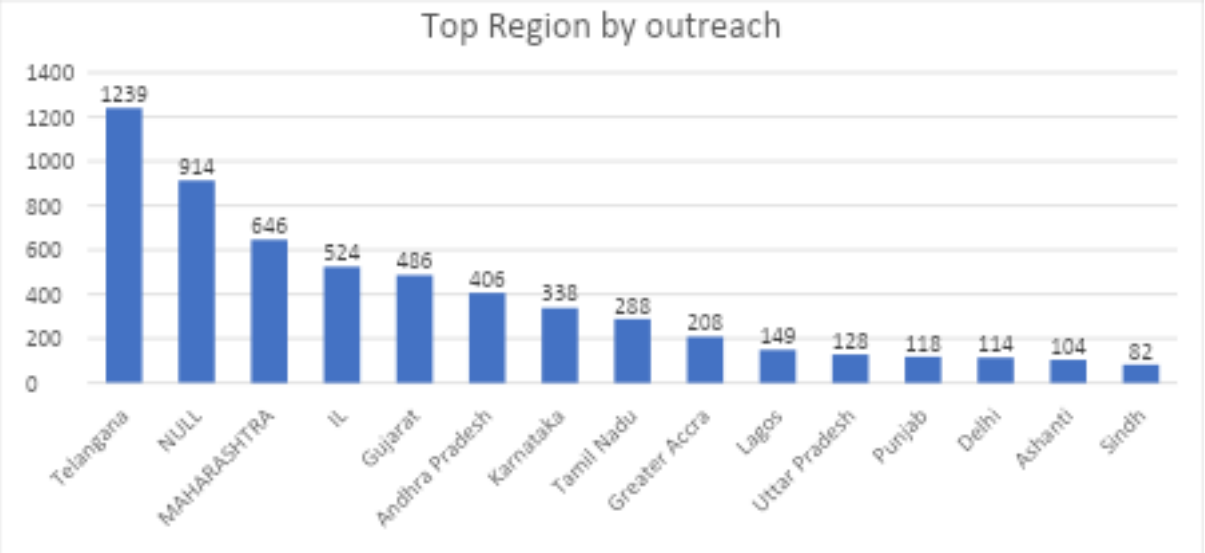
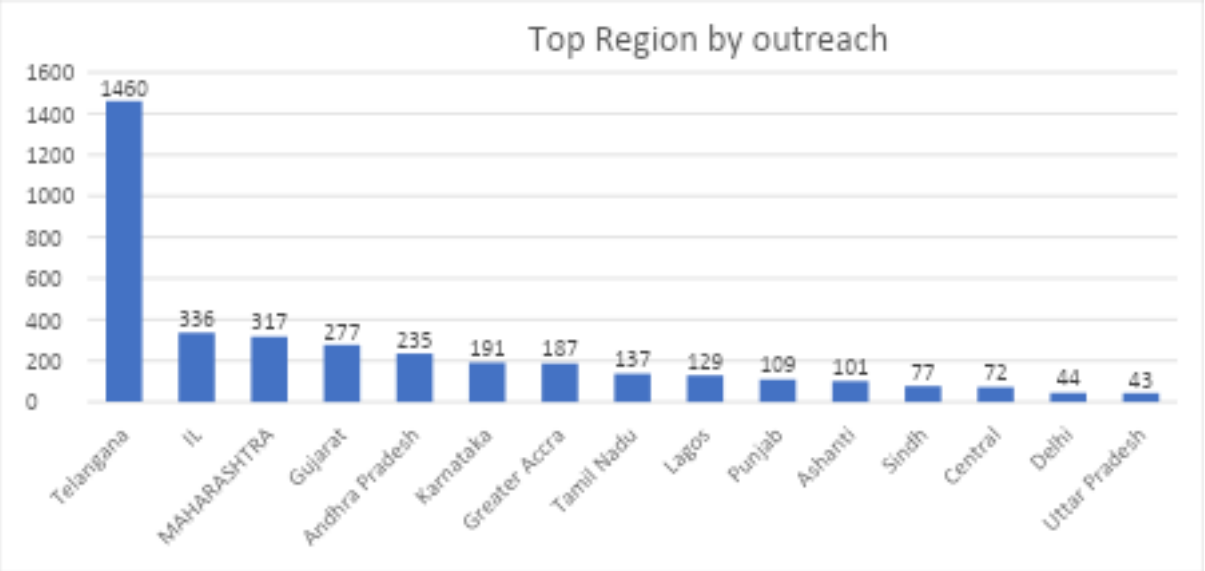


Before cleaning



After Cleaning



Appendix A: Jupyter Notebook – Data cleaning process

The complete Jupyter Notebook used for cleaning and preprocessing the dataset is attached. It contains all python codes, visualizations, and outputs that support this report.

File Name: Data cleaning file.ipynb

https://drive.google.com/file/d/1b8IiErJOXTbxf8K1ukjCF3FiCPl2QFdx/view?usp=drivesdk

Exported Version: Data cleaning file.pdf

https://drive.google.com/file/d/144agISVHb-yWA-rx88zrQ8Hd0i8TpWS5/view?usp=drivesdk

Exploratory Data Analysis & Dashboard Prototype
Dataset: DePaul Outreach Dataset

Introduction:

This exploratory analysis of the depaul outreach dataset highlights critical patterns in outreach effectiveness, engagement behavior and conversion outcomes. The findings reveal opportunities to refine targeting, timing, and follow-up strategies to maximize recruitment impact.

Part 1: Exploratory Data Analysis (EDA) Insights Report

Objective

The purpose of this analysis was to explore outreach and application data to uncover trends, performance gaps, and optimisation opportunities. The insights focus on outreach effectiveness, response behaviour, and conversion signals that align with recruitment and engagement goals.

Key Insight 1: Outreach Volume Is High, but Engagement Is Uneven

Finding

Outreach activity shows high overall volume; however, engagement outcomes (responses or positive results) are concentrated within a small portion of total contacts.

Supporting Evidence

- Majority of records fall under “attempted” or “no response” outcomes
- Fewer contacts progress to positive engagement or final conversion

Why This Matters

High outreach volume alone does not guarantee performance. This indicates an opportunity to improve **message relevance, timing, or targeting** rather than increasing volume.

Opportunity Identified

- Prioritise high-response segments instead of broad outreach

- Refine messaging for low-response groups

Key Insight 2: Multiple Contact Attempts Improve Response Likelihood

Finding
Records with more than one contact attempt show higher response or outcome completion rates compared to single-attempt contacts.

Supporting Evidence

- Engagement rates increase after the second or third attempt
- First-attempt outreach frequently results in no response

Why This Matters
This supports the value of **follow-up strategies** and structured outreach cadences instead of one-off contact attempts.

Opportunity Identified

- Set a minimum follow-up rule (e.g., 2–3 attempts per lead)
- Monitor diminishing returns beyond a certain attempt threshold

Key Insight 3: Performance Varies by University Segment and Region

Finding
Certain university groups and regions consistently outperform others in engagement and progression.

Supporting Evidence

- Some universities show higher response and application progress
- Other segments require significantly more attempts for similar outcomes

Why This Matters
Not all segments deliver equal value. Understanding this allows for **data-driven prioritisation** of outreach resources.

Opportunity Identified

- Focus outreach efforts on high-performing universities
- Customise messaging for underperforming regions

Key Insight 4: Timing Impacts Outreach Effectiveness

Finding
Engagement trends fluctuate over time, with noticeable peaks during specific periods.

Supporting Evidence

- Certain weeks/months show higher response rates
- Engagement drops during other periods despite consistent outreach volume

Why This Matters
Outreach success is sensitive to timing, academic calendars, and applicant readiness.

Opportunity Identified

- Align outreach campaigns with peak engagement periods
- Reduce low-yield outreach during inactive periods

Key Insight 5: Conversion Drop-Off Occurs Late in the Funnel

Finding
A measurable drop-off occurs between positive contact outcomes and final results (e.g., admissions or next-step completion).

Supporting Evidence

- Strong early engagement does not always translate into final outcomes
- Checklist and follow-up stages appear to be friction points

Why This Matters
Optimising mid-to-late funnel steps may yield better results than increasing initial outreach.

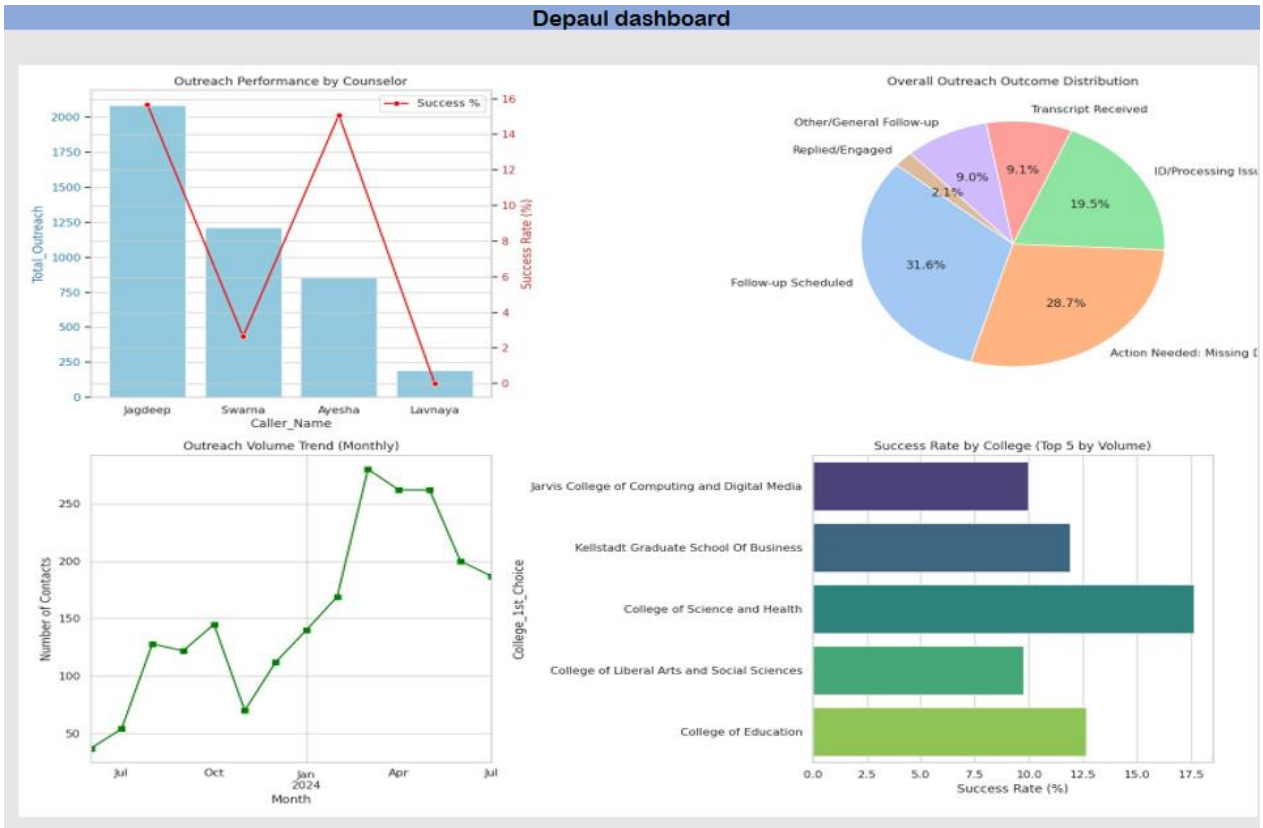
Opportunity Identified

- Improve follow-up communication after initial engagement

- Identify and address common blockers in the application process

Dashboard Prototype

This dashboard transforms outreach data into strategic insights, pinpointing top performers, process gaps and high-yield academic units for smarter engagement.



The above dashboard is designed to provide high-level KPI tracking while allowing for deep dives into counselor and college performance.

A four-chart dashboard shows:

1. Outreach Performance by Counselor

This is a dual-axis chart showing outreach volume (in bars) and success rate (in line) by counselors. The volume and conversion levels of Jagdeep and Ayesha are on the lead and this is an indicator of high efficiency. Swarna and Lavanya have lower success rates which indicates that coaching or reassigning them may be possible.

Why it is important: Helps recognize the high performers and deploy the counselors more effectively to ROI.

Dashboard Prototype

2. Overall Outreach Outcome Distribution

A pie chart depicting the results of outreach. The biggest slice - Follow-up Scheduled (31.6%) shows that there is great initial engagement, but there is a necessity to continue follow-through. Bottlenecks can be seen in terms of "Missing Documents (28.7%) and ID/Processing Issues (19.5%).

Why it is important: Exposes locations of outreach stalls and areas of automation or reminders that would enhance conversion.

3. Outreach Volume Trend (Monthly)

A line graph of the outreach activity in the period between July 2023 and July 2024. The volume increases in early 2024, which peaks around April-May, presumably in line with admission cycles. The fact that there is a slight dip in July indicates seasonal slackness.

Why it is important: It makes it possible to plan the strategies around the periods of high demand and resource optimization.

4. Success Rate by College (Top 5 by Volume)

A horizontal bar graph of the ranking of colleges in terms of outreach success. The College of Education and College of Science and Health are the best performing and Jarvis and Kellstadt have the lowest conversion.

Why it is important: Guides target strategy- high yield colleges to get the most out of the counselor.

Hence, the overall description is that this dashboard will turn raw outreach data into strategic information, for instance, who are the best counselors, what are the gaps in processes and what high-converting academic units should be prioritized in following up.

Pivot table Analysis:

Comments	Sum of ID	Sum of Campaign_Id	Sum of Reference_ID-2
⊞Call From Student	2808810	2808810	1066005974
⊞Call To Student	91674852	91674852	69656354311
⊞Complete Case	423702090	423702090	2.45178E+11
⊞Email From Student	16816704	16816704	33228331704
⊞Email Sent To Study Group Team	103190604	103190604	73515457092
⊞Email To Student	146649612	146649612	1.18948E+11
⊞Sent For Evaluation	2060471640	2060471640	1.82626E+12
⊞Whatsapp To Student	2312895	2312895	1363142097
Grand Total	2847627207	2847627207	2.36922E+12

Dashboard Prototype

Row Labels	Sum of Reference_ID-2
⊞College of Communication	36686582660
⊞College of Education	78756514412
⊞College of Education (Fully Online)	5613673352
⊞College of Liberal Arts and Social Sciences	80627047671
⊞College of Science and Health	1.05739E+11
⊞College of Science and Health (Fully Online)	473843950.4
⊞Jarvis College of Computing and Digital Media	1.14767E+12
⊞Jarvis College of Computing and Digital Media (Fully Online)	22480987893
⊞Kellstadt Graduate School Of Business	7.76171E+11
⊞Kellstadt Graduate School Of Business (Fully Online)	9105815348
⊞School of Continuing and Professional Studies	8467239769
⊞School of Continuing and Professional Studies (Fully Online)	2369219752
⊞School of Music	10224347536
⊞School of Nursing	11352840247
⊞School of Nursing (Fully Online)	473843950.4
⊞School of Psychology	30762500502
⊞School of Public Service	27849689097
⊞School of Public Service (Fully Online)	2559712973
⊞The Theatre School	11841396294
Grand Total	2.36922E+12

1. Interaction Type Breakdown

This pivot table categorizes student engagement by communication type—calls, emails, WhatsApp messages, evaluations and case completions. It quantifies outreach volume across each channel, helping identify which modes are most used and potentially most effective.

2. College-Level Engagement Summary

This pivot table aggregates student communication volume by college and program type (standard vs fully online). Jarvis College and Kellstadt Graduate School show high activity while niche schools like Music and Theatre also show strong engagement.

Actionable Recommendations:

- Counselor Deployment:

Focus on Jagdeep and Ayesha (high success) as well as coach Swarna and Lavanya.

- Document Collection:

Automate reminders to reduce 28.7% “missing docs” stalls.

- Follow-Up Strategy:

Strengthen second-touch outreach since 31.6% are stuck in follow-ups.

- College Targeting:

Prioritize Education and Science/Health (highest success) and refine scripts for low-yield colleges.